Subpixel Extraction of Small Water Bodies Using a Water Fraction-Water Index (WF-WI) Framework Based on Multispectral Imagery

Taixia Wu, Jiayun Zuo, Hongzhao Tang, Lilin Zhang, and Xuege Wang

***Abstract*—Small water bodies (SWBs) are critical components for maintaining regional ecological balance and hydrological cycles. However, their small size and fragmented distribution make them susceptible to mixed-pixel effects in medium- to low-resolution remote sensing imagery, resulting in significant omission errors in existing extraction methods. To achieve high-accuracy mapping of SWBs, this study proposes a coupled framework integrating water indices with subpixel decomposition-the water fractionwater index (WF-WI) framework. The framework implements a three-stage extraction process: first, potential water-containing mixed pixels are identified based on the bimodal characteristics of water indices; second, water and land endmembers are adaptively determined through a spectral angle matching algorithm that combines global statistics with local spatial information; finally, a WI-based pixel decomposition model estimates water fractions, followed by threshold segmentation to extract SWBs. Experiments were conducted in three typical regions: Rizhao City in Shandong Province, Jingmen City in Hubei Province, and Chuzhou City in Anhui Province, using Landsat 8/9 imagery from 2021 to 2022. Using modified normalized difference water index (MNDWI) as an example to instantiate the WF-WI framework (denoted as WF-MNDWI), results demonstrate that the method significantly outperforms support vector machines (SVMs), random forests (RFs), the Joint Research Center (JRC) global surface water dataset, and traditional index-based methods in overall accuracy (OA) and $F1$ scores, with $F1$ values of 0.9818, 0.9122, and 0.9619, respectively. The framework exhibits excellent generality and transferability, supporting multiple water indices including MNDWI, NDWI, and automated water extraction index (AWEI) (instantiable as WF-MNDWI, WF-NDWI, and WF-AWEI), and successfully transferring to Sentinel-2 imagery, substantially improving SWB detection across different sensor platforms. Specifically, WF-MNDWI is robust in most scenarios and resistant to topographic shadow interference; WF-AWEI shows superior performance in extracting narrow rivers; although WF-NDWI is affected by highly reflective surfaces, it still outperforms traditional indices and possesses significant application potential due to its band generality. The proposed WF-WI framework enhances subpixel water detection through a dual-threshold pre-classification and spatially adaptive end-member selection strategy, while focusing on "potential water-containing mixed pixels" to optimize computational efficiency. This study provides an innovative solution for high-accuracy extraction of SWBs in mixed-pixel environments and establishes a methodological foundation for large-scale, multisource remote sensing monitoring of dynamic water bodies.**

***Index Terms*—Pixel decomposition, small water bodies (SWBs), subpixel extraction, water fraction, water index (WI).**

I. Introduction

SMALL water bodies (SWBs), as a vital component of the global inland surface water system, play a unique and irreplaceable role in ecosystems [1], [2], [3]. SWBs, including ponds, small lakes, and ditches of either artificial or natural origin, constitute the most numerous freshwater resources worldwide [4]. Lakes and ponds smaller than 10 hectares account for approximately 30% of the global freshwater surface area [5]. Area is one of the key factors determining the ecological characteristics of SWBs [6]; however, no universally accepted scientific or legislative definition currently exists [4]. In relevant studies, SWBs are classified as either permanent or seasonal, typically ranging in size from less than 1 hectare to less than 50 hectares [7], [8], [9]. SWBs not only provide essential water resources for agricultural irrigation and aquaculture [10], [11] but also exhibit nitrogen removal capacities comparable to those of large water bodies [12], [13]. In addition, by promoting infiltration and temporarily storing water, SWBs can intercept stormwater runoff and locally mitigate flooding in a distributed manner [8], [14]. In specific regions, such as peat wetlands, SWBs exhibit a per-unit-area carbon sequestration capacity that far exceeds that of forests or large lakes [15]. SWBs are also critical for maintaining freshwater biodiversity [16], [17]. With equivalent total area, SWBs support a greater diversity of freshwater species than large water bodies [12] and have become essential refuges for many endangered species [4]. However, due to their small scale, high density, and widespread distribution [18], SWBs are increasingly threatened by both climate change and human activities, with these threats intensifying over time. As a result, their eco-system functions are being severely disrupted [19]. An analysis of data from China between 1995 and 2015 revealed a 71% reduction in SWB area [20]. Despite growing recognition of their importance, significant gaps remain in the spatiotemporal monitoring of SWBs [21], primarily due

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to technical challenges arising from their subpixel spatial characteristics and high dynamism.

Remote sensing has been proven to be a cost-effective tool for surface water extraction [22], [23], [24]. Various remote sensing sensors with different spectral, spatial, and temporal resolutions have been utilized in surface water studies [25], [26], [27]. However, existing methods still face significant limitations in the identification of SWBs. Main-stream water extraction methods can be broadly categorized into supervised classification and water index-based approaches. The former relies on complex classifiers such as support vector machine (SVM) [28] and random forest (RF) [29], whose performance is constrained by the quality of training samples and model generalization capabilities. The latter constructs various index models by analyzing the spectral characteristics of water and background surfaces and selecting relevant bands [30], [31], followed by water extraction based on a specified threshold. Examples of commonly used water indices include the normalized difference WI (NDWI) [32], modified NDWI(MNDWI) [33], and the automated water extraction index (AWEI) [34]. However, due to the small size of SWBs and their weak spectral response, they are difficult to detect in medium- and low-resolution remote sensing imagery [13]. For instance, although the Landsat series of satellites has been widely used for surface water mapping owing to its high-quality, detailed, and continuous data records [29], [35], [36], it is noteworthy that approximately 10.9% of global inland surface water appears as mixed pixels in Landsat imagery [37].

In recent years, researchers have attempted to improve water detection using higher-resolution Sentinel-2 imagery (10–20 m). For instance, Farhadi et al. [38] proposed novel water indices such as the flood/water extraction index (FWEI) and the Sentinel-2 flood mapping index (SFMI) [39]. The FWEI index, which combines the mean of visible bands (blue, green, red) with the near-infrared (NIR) band, effectively enhances the extraction of both permanent water bodies and floodwaters, demonstrating strong robustness in complex environments [38]. The SFMI index enables water mapping without relying on preset thresholds and achieves high accuracy in flood monitoring scenarios [39]. By leveraging the 10 m resolution bands of Sentinel-2, these indices represent significant technical advancements in detecting SWBs such as fish ponds and narrow rivers. However, despite these improvements, including FWEI and SFMI, these indices are fundamentally pixel-level analysis methods. They remain limited in addressing a core challenge of medium-resolution remote sensing imagery: the mixed-pixel problem. When the water fraction within a pixel is low, these indices often fail to detect subpixel water information, resulting in omission errors.

Given the small size of SWBs, subpixel extraction techniques based on mixed pixel decomposition offer an effective approach for their fine-scale identification. Mixed pixel decomposition is commonly applied in hyperspectral remote sensing imagery [40], [41], treating the spectral characteristics of pure pixels as endmembers and considering mixed pixels as linear or nonlinear combinations of these endmembers [42], [43]. This technique enables the retrieval of subpixel endmember fraction information and is widely used for high-resolution land cover mapping [44], [45]. It has been successfully applied to various water environment information extraction tasks, such as urban water bodies extraction [46], reservoir surface area estimation [25], and dynamic monitoring of surface water [35], [47]. The selection of end-members directly affects the accuracy of mixed pixel decomposition. Due to the intraclass heterogeneity and interclass similarity of land surface spectral features [48], [49], traditional methods based on the "purest pixel" criterion do not always yield optimal results [50]. To mitigate the impact of endmember variability in mixed pixel decomposition, some researchers have proposed constructing unmixing models by exhaustively combining endmembers and selecting the optimal model based on the lowest error metric [51]. Although this approach guarantees a theoretically optimal solution, it faces a bottleneck in computational efficiency, limiting its practical applicability. A more feasible strategy is to enhance interclass variability and reduce intraclass variability from the spectral dimension. To enhance interclass variability, one approach is to select highly variable specific bands based on class separability, such as the use of shortwave infrared (SWIR) bands for soil–vegetation separation [52]. Additionally, strategies such as selecting stable band subsets [53], applying mean normalization [54], and spectral weighting [55] have been employed to suppress intraclass variability.

Compared to the high-cost hyperspectral remote sensing imagery [56], this study proposes a coupled framework model, the water fraction—water index (WF-WI) framework, based on commonly available multispectral data. This framework significantly improves the extraction efficiency and accuracy of SWBs at the subpixel scale by innovatively combining the feature enhancement mechanism of water indices with an optimized endmember selection strategy. Specifically, the water index (WI) highlights the core spectral features of water bodies through band ratio operations, effectively suppressing intraclass variability factors such as suspended sediment concentration, chlorophyll content, and water depth, while also reducing the impact of lighting conditions and atmospheric interference. This results in a set of water endmembers with higher spectral consistency. To address the low computational efficiency of traditional mixed pixel decomposition methods when processing global pixels, this study focuses on potential water-containing mixed pixels between the main and secondary peaks in the histogram of WI's bimodal distribution (with the primary peak representing pure water and the secondary peak representing nonwater land cover). A dual-threshold segmentation algorithm [57] is applied to identify these mixed pixels. By incorporating the theory of spatial dependence [58], an adaptive water endmember selection method based on spatial neighborhood (5×5 window) is established, effectively addressing the spatial heterogeneity challenges of SWBs. The selection of land endmembers is determined through the spectral angle matching algorithm to identify the dominant land type, a strategy particularly suited for spectral unmixing of mixed pixels with low water fraction.

In summary, current remote sensing methods for SWBs extraction face a fundamental challenge. Supervised classification methods (e.g., SVM, RF) heavily depend on sample quality and exhibit limited generalization; global open datasets

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[e.g., Joint Research Centre (JRC)] often lack sufficient spatial detail; and traditional threshold-based water indices (e.g., MNDWI, NDWI), while computationally efficient, are constrained by the mixed-pixel problem and are unable to detect subpixel water bodies. Although recent studies have proposed higher-performance indices using Sentinel-2 data, such as FWEI and SFMI, these methods remain fundamentally pixel-level analyses and cannot fundamentally resolve omissions caused by mixed pixels. Therefore, there is an urgent need for a novel approach that integrates the advantages of WI features with the fine-scale quantification capability of mixed-pixel decomposition. Such a method should intelligently focus on the most critical mixed-pixel regions rather than performing global computations, thereby achieving a balance between accuracy and efficiency. The WF-WI framework proposed in this study is specifically designed to address this challenge.

Therefore, this study is centered around three core objectives: 1) to develop a dual-threshold segmentation model based on the WI for the accurate identification of potential water-containing mixed pixels; 2) to develop a spatial neighborhood-driven water endmember selection algorithm and a land endmember extraction method based on spectral matching algorithms, establishing an optimized remote sensing subpixel decomposition framework for WI; and 3) to construct the WF-WI method framework based on Landsat 8/9 multispectral data for the subpixel scale fine extraction of SWBs. This method optimizes the allocation of computational resources by focusing on key mixed regions, significantly reducing the computational load of global unmixing in traditional mixed pixel decomposition. It provides a new technological pathway for the application of multispectral remote sensing data in SWB monitoring.

II. Study Area and Data

A. Study Area

Based on the spatial density, morphological complexity, and underlying surface environmental heterogeneity of SWBs, three study areas in China were selected to validate the effectiveness of the framework proposed in this study (Fig. 1). Study Area A (Northwest Rizhao City, Shandong Province): The Qingfengling Reservoir serves as the primary water body, with surrounding SWBs (such as storage ponds, narrow channels, and irrigation ponds) sparsely distributed. The underlying surface exhibits significant uniformity, primarily consisting of bare land and fallow farmland. Study Area B (Southwest Jingmen City, Hubei Province): This area exhibits a high-density clustered distribution of SWBs (including agricultural reservoirs, irrigation channels, and narrow rivers), located in a complex rural landscape with highly diverse land use types and strong human agricultural activity interference. Study Area C (Eastern Chuzhou City, Anhui Province): This area includes large-scale fishponds and numerous small to medium-sized reservoirs. Some regions contain high-density urbanized areas, forming a natural-artificial composite distribution pattern of SWBs, with prominent spectral mixing effects from the underlying surface.

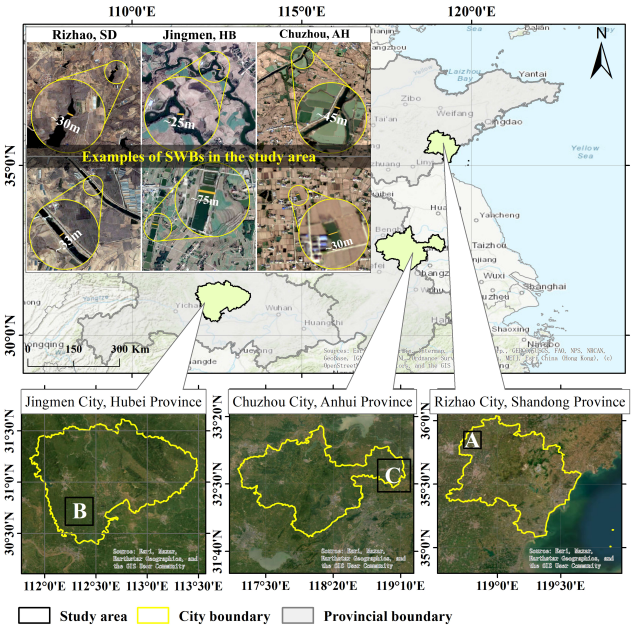

Fig. 1. Location of the study area and characteristics of typical SWBs.

TABLE I
Study Area Image Data and Verification Sample Information. L8: Landsat 8; L9: Landsat 9; S2: Sentinel-2; GE: Google Earth; Res.: Spatial Resolution; No.: Number of Samples

Study area	Source	Date	Res. (m)	Sample type	No.
Rizhao	L9	2022-03-28	30	Water	495
	S2	2022-03-27	10	Non-Water	505
	GE	2022-03-29	<2.5		
Jingmen	L8	2021-01-01	30	Water	381
	S2	2021-01-02	10	Non-Water	619
	GE	2021-01-18	<2.5		
Chuzhou	L9	2021-11-24	30	Water	485
	S2	2021-12-02	10	Non-Water	515
	GE	2021-11-15	<2.5		

B. Data Sources

This study utilized the global human settlement layer (GHSL) 10 m resolution built-up dataset (Dataset ID: JRC/GHSL/P2023A/GHS_BUILT_S_10m/2018) released by the JRC of the European Commission to extract built-up areas in the study regions and construct a building mask. The mask was resampled to 30 m resolution via the Google Earth Engine (GEE) platform to achieve spatial consistency with the Landsat imagery.

Remote sensing imagery was selected from the Landsat-8/9 Level-2 Surface Reflectance products jointly provided by NASA and the U.S. Geological Survey (USGS). Image selection was based on the following criteria: 1) temporal matching, giving priority to images closest in acquisition date to available high-resolution validation data (Sentinel-2 and Google Earth) to ensure validation reliability; 2) data availability, selecting either Landsat-8 or Landsat-9 images according to data availability for each study area while satisfying temporal matching (see Table I); and 3) cloud coverage, screening for cloud-free images over the study areas between 2021 and

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2022 (cloud cover <5%). Using the GEE platform, all images underwent radiometric calibration, atmospheric correction, and cloud masking. To focus on water spectral characteristics, only visible to SWIR bands (B1–B7) were retained, while low signal-to-noise ratio thermal infrared bands were excluded.

Landsat imagery was selected for validating the proposed framework rather than higher-resolution Sentinel-2 data based on the following considerations: 1) fairness and reproduceibility of comparison: this study aims to systematically compare the proposed method with a series of classical Landsat-based approaches (e.g., MNDWI, SVM, RF) and global open water datasets (JRC). Using the same data source ensures the fairness of the comparison and the reliability of the conclusions; 2) spectral response consistency: all Landsat bands (visible to SWIR) share a 30 m spatial resolution, ensuring spectral homogeneity within pixels, which provide a more stable theoretical basis for the linear mixing model in subpixel decomposition; and 3) universality validation of the method: we first aim to verify the effectiveness of the WF-WI framework on a platform with consistent spectral characteristics and a mature algorithmic ecosystem, laying a clearer and more reliable foundation for subsequent application of the framework to other multispectral data platforms such as Sentinel-2.

Accuracy assessment was conducted using contemporaneous high-resolution Sentinel-2 imagery and Google Earth imagery as reference data. The specific image acquisition dates, spatial resolutions, and the number of validation sample points for each study area are detailed in Table I.

III. Methods

A. WF-WI Framework

In this study, a coupled framework model, termed the WF-WI Framework, is proposed to achieve accurate extraction of SWBs through the fusion of WI and mixed pixel decomposition models. As shown in Fig. 2, the WF-WI framework extracts water bodies through the following steps.

1) *Initial Classification of Pixel Categories:* Based on the bimodal statistical characteristics of the WI histogram, and using automatically derived high- and low-confidence thresholds, pixels are classified into three categories: "confirmed water," "confirmed nonwater," and "potential water."
2) *Determination of Endmembers:* For pixels categorized as "potential water," a dual-constraint mechanism based on spatial dependence and spectral characteristics is employed. Specifically, A moving window approach, along with the minimum spectral angle mapping technique, is used to determine the water and land endmember combinations for these pixels.
3) *Calculation of Water Fraction:* For "potential water" pixels, a WI-based pixel decomposition model is applied to estimate the water fraction, while "confirmed nonwater" and "confirmed water" pixels are assigned values of 0 and 1, respectively.
4) *Water Bodies Extraction:* Water bodies are then extracted by applying an appropriate threshold. Different water indices can be adapted to various underlying surface backgrounds, and the WF-WI framework is compatible with multiple water indices. In this study, the WF-WI framework was instantiated using the MNDWI index (denoted as WF-MNDWI) for water extraction experiments. In subsequent discussions, NDWI and AWEI_{nsh} (hereafter uniformly abbreviated as AWEI) were also used to instantiate the framework as WF-NDWI and WF-AWEI, respectively, to assess the generality and applicability of the framework.

B. Initial Classification of Pixel Categories

When the histogram of a WI exhibits a bimodal distribution, the main peak corresponds to pure water, while the secondary peak represents nonwater land cover types. The transitional region between the two peaks mainly includes two types of targets: subpixel water bodies and spectrally confused features. Traditional single-threshold methods have inherent limitations: a high threshold reduces the misclassification of nonwater surfaces but leads to omission of subpixel water bodies, whereas a low threshold facilitates the detection of subpixel water bodies but simultaneously increases the misclassification of land features. Due to the environmental complexity and the subpixel characteristics of SWBs, the use of a single threshold often fails to achieve satisfactory extraction results. Therefore, this study adopts a dual-threshold segmentation strategy to enable hierarchical extraction of water bodies information.

Water indices effectively enhance the spectral distinction between water bodies and land surfaces through specific band combination operations. Given that pure water bodies exhibit higher WI values, this study employs a high-confidence threshold ($T_{\text{pure-water}}$) to extract a set of "confirmed water" pixels, denoted as C_{water} [see (1)]. Conversely, a low-confidence threshold ($T_{\text{non-water}}$) is applied to exclude land pixels with lower WI values, corresponding to "confirmed nonwater," denoted as $C_{\text{non-water}}$ [see (2)]. The transitional region between these two thresholds contains subpixel SWBs, which are regarded as "potential water," denoted as P_{water} [see (3)]

$$C_{\text{water}} = \left\{ i \in V \,|\, \text{WI}_i \geq T_{\text{pure-water}} \right\} \tag{1}$$

$$C_{\text{non-water}} = \left\{ i \in V \,|\, \text{WI}_i < T_{\text{non-water}} \right\} \tag{2}$$

$$P_{\text{water}} = \left\{ i \in V \,|\, T_{\text{non-water}} \leq \text{WI}_i < T_{\text{pure-water}} \right\} \tag{3}$$

where V represents all pixels in the image, and WI_i is the WI value of pixel i.

To obtain a bimodal WI histogram as effectively as possible, the following strategy was employed in this study: first, a loose threshold was applied to the WI to extract approximate water boundaries; next, the Canny edge detection algorithm was used to delineate these boundaries, and several pixels were buffered both inside and outside the boundaries to form a buffer zone. Only the WI values within this buffer zone were extracted, and a histogram was constructed based on these values. Finally, thresholds were determined using the "buffered WI histogram."

The automatically determined high-confidence and low-confidence thresholds are calculated using (4) and (5). This design is based on the following considerations: compared

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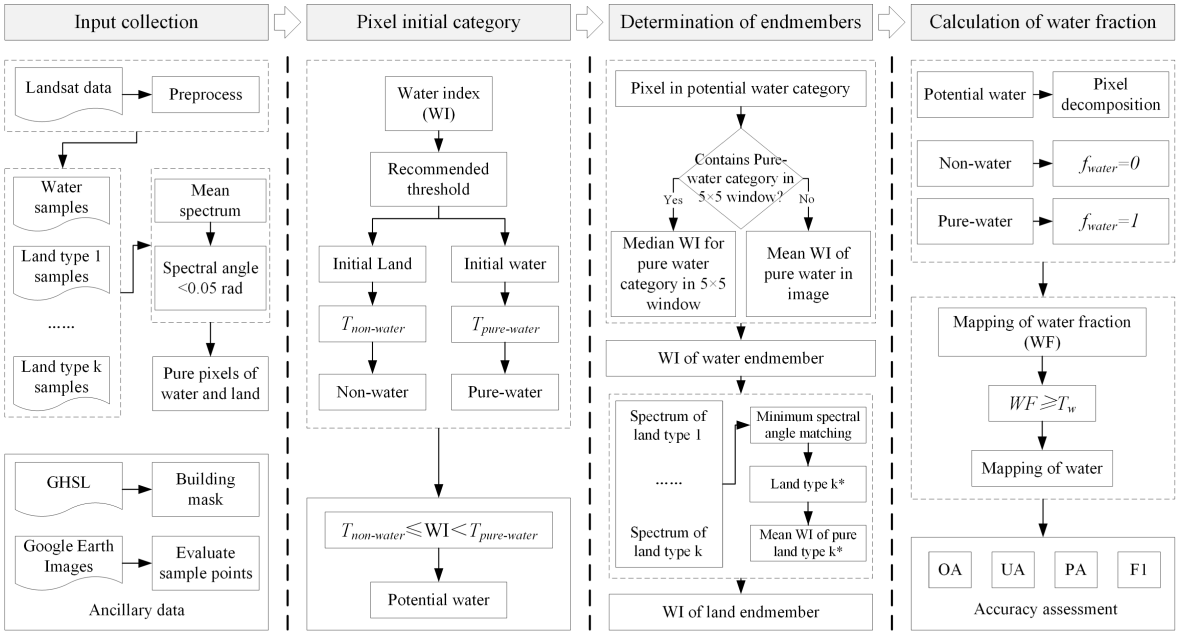

Fig. 2. Technical flowchart of WF-WI framework.

TABLE II
WI CALCULATION FORMULA AND RELATED INFORMATION

Index	Equation	Threshold	Source
MNDWI	$(G - \text{SWIR1})/(G + \text{SWIR1})$	0	[33]
NDWI	$(G - \text{NIR})/(G + \text{NIR})$	0	[32]
AWEI_{nsh}	$4 \times (G - SWIR1) - (0.25 \times NIR + 2.75 \times SWIR2)$	0	[34]

with the mean, the use of the median effectively suppresses the influence of outliers; in addition, the high- and low-confidence thresholds are obtained through positive and negative offsets, respectively, ensuring both the accuracy of the C_{water} category and the retention of potential water-containing mixed pixels (P_{water}):

$$T_{\text{pure-water}} = \max\left\{T_{\text{initial}}, \frac{M_1 + \text{std}_1}{2}\right\} \tag{4}$$

$$T_{\text{non-water}} = \min\left\{T_{\text{initial}}, \frac{M_2 - \text{std}_2}{2}\right\}. \tag{5}$$

In these equations, T_{initial} represents the initial segmentation threshold of the WI, which is set using the recommended threshold corresponding to the specific WI (see Table II). M_1/std_1 and M_2/std_2 denote the median and standard deviation of the WI values above and below T_{initial}, respectively.

C. WI-Based Pixel Decomposition Model

In this study, the acquisition of endmember fraction information is based on the linear mixing model from the mixed pixel decomposition method. During mixed pixel decomposition, dimensionality reduction is usually necessary, as certain correlations exist among the spectral bands of remote sensing imagery, making it unnecessary for all bands to participate in the analysis.

Given the construction principles of water indices, this study treats the WI as a form of dimensionality reduction and feature extraction from the multispectral data of remote sensing images. Therefore, pixel decomposition is performed directly based on the WI. Regarding the determination of endmembers, two types of endmembers are considered: the water end member and the land endmember. The pixel decomposition model based on the WI is formulated as follows:

$$\begin{cases} \text{WI}_{\text{water}} f_{\text{water}} + \text{WI}_{\text{land}} f_{\text{land}} = \text{WI} \\ f_{\text{water}} + f_{\text{land}} = 1. \end{cases} \tag{6}$$

In this equation, WI_{water} and WI_{land} represent the WI values of pure water and pure land endmembers, respectively. f_{water} and f_{land} denote the fractional proportions of water and land components, respectively, which sum to 1. WI denotes the WI value of the pixel.

Before calculating the water fraction, publicly available building datasets are used to mask the building areas in the imagery, and the water fraction within these areas is directly assigned a value of 0. For pixels classified as P_{water}, after determining the corresponding water and land endmembers, the f_{water} can be obtained through a simple transformation [see (7)]. For pixels categorized as C_{water} and C_{nonwater}, the water fractions are directly assigned values of 1 and 0, respectively,

$$f_{\text{water}} = \frac{\text{WI} - \text{WI}_{\text{land}}}{\text{WI}_{\text{water}} - \text{WI}_{\text{land}}}. \tag{7}$$

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D. Determination of Endmembers

1) Determination of Reference Endmembers: In the process of mixed pixel decomposition, the selection of appropriate endmembers is crucial. Considering the characteristics of SWBs and land cover types, this study adopts a differentiated strategy for endmember determination. Due to the lack of uniform and stable spectral characteristics for the building class, publicly available building datasets are first used to mask building areas within the study region. Subsequently, with the aid of high-resolution Google Earth and Sentinel-2 imagery, a set of representative surface samples was manually selected for each study area, ensuring that the sample types covered the main land cover classes apart from built-up areas. The primary types typically included water, vegetation, bare soil, and widely distributed unknown soil types. Several pure water pixels and multiple pure pixels of different land cover types were selected as typical land cover samples for the corresponding study area. The mean reflectance of each sample type across all bands was calculated to obtain representative reference spectra for the study area. Pixels whose spectral angles with the reference spectra were within 0.05 rad [59], [60] were considered as "pure pixels" of the corresponding land cover type. The mean WI values of these "pure pixels" are then computed to derive the reference water endmember value $\mathrm{WI}_{ref-water}$ and the reference land endmember values for different land types, denoted as $\mathrm{WI}_{ref-land1}, \mathrm{WI}_{ref-land2}, \ldots\ldots, \mathrm{WI}_{ref-landk}$.

2) Determination of Water Endmembers: In the P_{water} category, many SWBs are located at the water–land boundary, or there are pure water bodies nearby. Based on the spatial autocorrelation theory of the first law of geography, the true water endmember characteristics of these pixels exhibit higher spatial dependence with the surrounding pure water bodies. Therefore, in this study, when selecting the water endmember, we prioritize the consideration of neighboring spatial information. Specifically, for each pixel, all pixels within a 5×5 surrounding window are first sampled, and pixels classified as C_{water} are considered as candidate water endmembers. The median of their WI values is taken as the water endmember for that pixel. If no C_{water} pixels are found within the 5×5 window, the reference water endmember $\mathrm{WI}_{ref-water}$ is assigned as the water endmember for that pixel. The rule for determining the water endmember value $\mathrm{WI}_{\mathrm{water}i}$ for each pixel i in the P_{water} category is as follows:

$$\mathrm{WI}_{\mathrm{water}\ i} = \begin{cases} \mathrm{Median}\left\{\mathrm{WI}_j \mid j \in (N_i \cap C_{\mathrm{water}})\right\}, if|j| \neq 0 \\ \mathrm{WI}_{ref-water},\ if|j| = 0. \end{cases} \tag{8}$$

In this equation, N_i represents all the neighboring pixels within the 5×5 window surrounding pixel i, C_{water} denotes the set of "confirmed water" pixels, $N_i \cap C_{\mathrm{water}}$ represents the set of "confirmed water" pixels within the 5×5 neighborhood of pixel i, $|j|$ denotes the number of elements in the set j, WI_j represents the WI value of each pixel j in the set $N_i \cap C_{\mathrm{water}}$, and $\mathrm{Median}\{\mathrm{WI}_j \mid j \in (N_i \cap C_{\mathrm{water}})\}$ denotes the median of the WI values corresponding to all pixels in the set j, while $\mathrm{WI}_{ref-water}$ is the reference water endmember value.

3) Determination of Land Endmembers: In this study, mixed pixels are simplified to consist of two components: water and land. Traditional pixel binary models typically assume the land endmember as a fixed value. However, in real remote sensing scenarios, land cover types exhibit complex diversity, and significant spectral differences exist between different surface types. According to the theory of the linear mixing model, when the water proportion in a pixel is low, its spectral characteristics are primarily influenced by the dominant land type. Based on reference spectra of land types, the minimum spectral angle matching algorithm can effectively identify the dominant land type in water–land mixed pixels. Therefore, the method for determining the land endmember involves calculating the spectral angle between the pixel spectrum and the reference land spectra for each land type. The land type k^* with the smallest spectral angle relative to the mixed pixel spectrum is selected, and the corresponding reference land endmember value $\mathrm{WI}_{ref-landk^*}$ is assigned as the land endmember for that pixel. For each pixel i in the P_{water} category, the land endmember value $\mathrm{WI}_{\mathrm{land}i}$ is calculated as follows:

$$\mathrm{WI}_{\mathrm{land}\ i} = \mathrm{WI}_{ref-landk^*}. \tag{9}$$

$$k^* = \mathrm{argmin}_k \left(\arccos \left(\frac{\sum (R_i \cdot R_{\mathrm{land}\ k})}{\sqrt{\sum R_i^2} \cdot \sqrt{\sum R_{\mathrm{land}k}^2}} \right) \right). \tag{10}$$

In this equation, R_i is the reflectance vector of pixel i, $R_{\mathrm{land}k}$ is the reference spectral vector of land type k, and k^* is the land type index that minimizes the spectral angle.

E. Accuracy Assessment

To quantitatively assess the accuracy of water body extraction, this study adopted a stratified random sampling strategy to allocate evaluation sample points, thereby ensuring a balanced representation between the "water" and "nonwater" classes while covering the entire study area. The specific procedure was as follows: First, based on high-resolution Sentinel-2 imagery acquired for each study area during the same period, the water and nonwater regions were preliminarily delineated using the WI method, serving as the basis for the two sampling strata. Subsequently, 500 candidate points were randomly generated within each stratum, resulting in an initial sample set comprising 1000 points.

The ground truth of all sample points was determined by visual interpretation of contemporaneous high-resolution imagery (Google Earth and Sentinel-2) using the "water-containing pixel criterion." Specifically, if any identifiable water body was present within the 30×30 m Landsat pixel where the sample point was located, the point was classified as "water"; otherwise, if no discernible water body existed, it was classified as "nonwater." The spatial distribution of validation sample points across the three study areas is shown in Fig. 3, while Table I presents the final counts of water and nonwater samples determined by interpretation for each study area.

Based on these sample points, the true positives (TPs), true negatives (TNs), false positives (FPs), and false negatives

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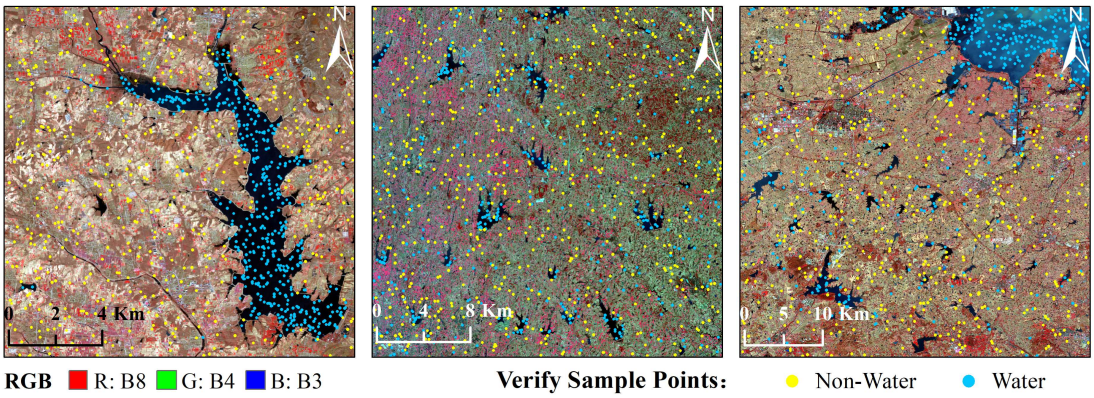

Fig. 3. Spatial distribution of the validation samples for accuracy assessment of water extraction.

(FNs) were calculated. Subsequently, four metrics—overall accuracy (OA), user's accuracy (UA), producer's accuracy (PA), and the $F1$ score—were employed to comprehensively evaluate classification performance.

OA reflects the overall agreement between classification results and actual land cover types, with higher values indicating greater accuracy. UA measures the reliability of a given class in the classification results, whereas PA quantifies the completeness of correctly classified instances for that class. The $F1$ score, as the harmonic mean of UA and PA, balances the trade-off between the two, with higher values indicating better overall classification performance.

IV. Results

A. Parameter Sensitivity Analysis and Optimal Threshold Determination

The threshold parameter T_w serves as the key parameter in the WF-WI framework for converting continuous water fraction values into the final binary water extraction results. This parameter represents the minimum water fraction value required for the model to effectively identify water pixels, that is, pixels with fraction values greater than or equal to T_w are classified as water.

Theoretically, the higher the model accuracy, the closer the optimal value of T_w should be to zero. However, in practical applications, uncertainties in the linear spectral unmixing model and noise inherent in remote sensing imagery introduce errors into the computed water fraction values. A too-low T_w threshold would lead to numerous FPs, whereas a too-high threshold would result in excessive FNs. Therefore, determining an optimal T_w is critical for balancing UA and PA, thereby maximizing OA.

To determine the optimal T_w, we conducted a sensitivity analysis over the interval [0.05, 0.95] with a step size of 0.05, evaluating the OA corresponding to each T_w value. Fig. 4 illustrates the relationships between T_w and OA for the WF-MNDWI, WF-NDWI, and WF-AWEI methods, which were instantiated using the MNDWI, NDWI, and AWEI, respecttively.

As shown in Fig. 4, the OA values generally exhibit an increasing (or stable) trend with increasing T_w, reaching a

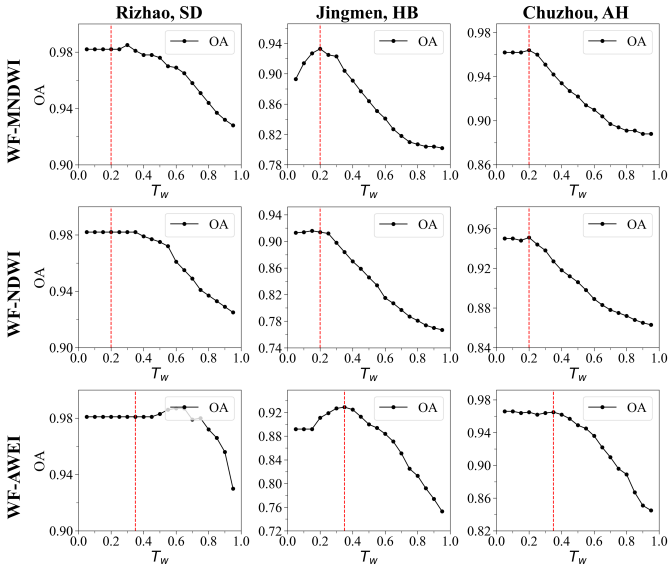

Fig. 4. Sensitivity analysis for parameter $\mathbf{T_w}$ in WF-WI.

peak before gradually declining. This pattern can be explained as follows: at lower threshold values, model errors cause numerous nonwater pixels to be incorrectly classified as water, leading to a low precision and suboptimal OA. As T_w increases, precision improves, resulting in higher OA values. However, once T_w exceeds a critical value, the gains in precision can no longer compensate for the substantial decline in recall—especially due to the omission of mixed pixels—leading to a decrease in OA.

Fig. 4 also reveals that the optimal T_w varies depending on the WI used. For WF-MNDWI, the optimal or near-optimal T_w values for the three study areas are 0.30, 0.20, and 0.20, respectively. For WF-NDWI the values are 0.35, 0.15, and 0.20, while for WF-AWEI, the values are 0.60, 0.35, and 0.35, respectively.

Finally, to balance performance with stability across different scenarios, we determined the following recommended thresholds: 0.20 for WF-MNDWI, 0.20 for WF-NDWI, and 0.35 for WF-AWEI (indicated by the red dashed lines in Fig. 4). These thresholds fall within the relatively flat "plateau"

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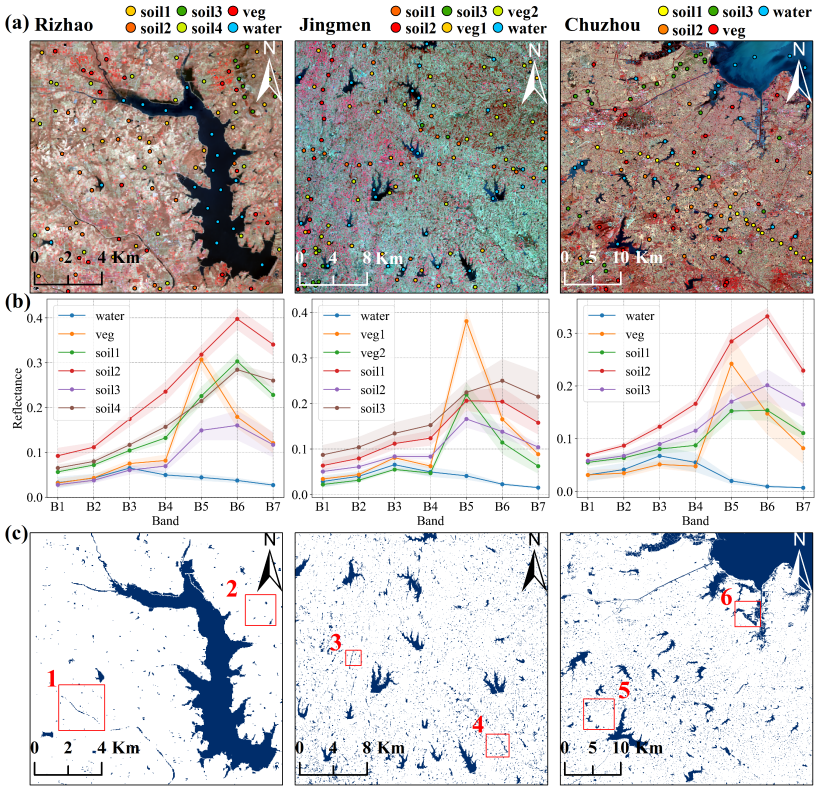

Fig. 5. Land cover spectral characteristics and water mapping results of WF-MNDWI. (a) Spatial distribution of sample points. (b) Mean spectral curves of different land cover types. (c) Water map over the study area.

regions of the OA curves rather than at sharp peaks, suggesting greater robustness to slight variations in input data or study areas, thereby ensuring stable generalization performance.

B. Water Body Mapping Results and Qualitative Comparison

To systematically evaluate the performance of the proposed framework, we take WF-MNDWI as an example to present the water extraction results across the entire study area and conduct a detailed qualitative comparison with other methods in selected representative subregions. The methods included for comparison comprise two supervised classification algorithms—SVM and RF, the globally available JRC water dataset, and three traditional WI-based methods: MNDWI, NDWI, and AWEI.

Fig. 5(a) shows the spatial distribution of typical pure land-cover samples across the study areas, while Fig. 5(b) presents the mean spectral reflectance curves for each land-cover class derived from these samples. These samples also serve as the training data for SVM and RF classifiers, with the specific sample sizes listed in Table III. Fig. 5(c) illustrates the water extraction results of WF-MNDWI at the recommended threshold (T_w = 0.20). Fig. 6 displays the water extraction results of all six comparative methods, where the JRC dataset products closest in acquisition time to the study area imagery were used, and each WI-based method was applied using its respective recommended threshold.

To further evaluate the qualitative performance, six representative subregions from the three study areas were selected

TABLE III
STATISTICS OF LAND COVER SAMPLE QUANTITIES USED FOR SUPERVISED CLASSIFICATION IN EACH STUDY AREA

Rizhao, SD		Jingmen, HB		Chuzhou, AH	
Type	Quantity	Type	Quantity	Type	Quantity
water	20	water	25	water	25
veg	20	veg1	20	veg	20
soil1	20	veg2	25	soil1	20
soil2	20	soil1	20	soil2	20
soil3	8	soil2	15	soil3	25
soil4	20	soil3	12	-	-

and are highlighted in red in Fig. 5(c). Enlarged local visualizations of the extraction results from all methods are shown in Fig. 7. The results demonstrate that in subregion 1, RF, JRC, and the three WI-based methods failed to effectively extract the narrow river channels, with only NDWI detecting a few scattered pixels; SVM identified some SWBs but with substantially lower completeness than WF-MNDWI. In subregion 2, apart from WF-MNDWI, the best-performing comparative method detected only two large water bodies on the right side. In subregions 3 and 4, JRC, NDWI, and AWEI performed worse than RF, while RF itself exhibited significant omission errors; MNDWI and SVM produced relatively better results but still underperformed compared with WF-MNDWI. In subregion 5, substantial differences in handling mountain shadows were observed: SVM misclassified large shadowed areas as water, RF exhibited similar FPs, and JRC produced fewer false detections but suffered from severe omissions; in

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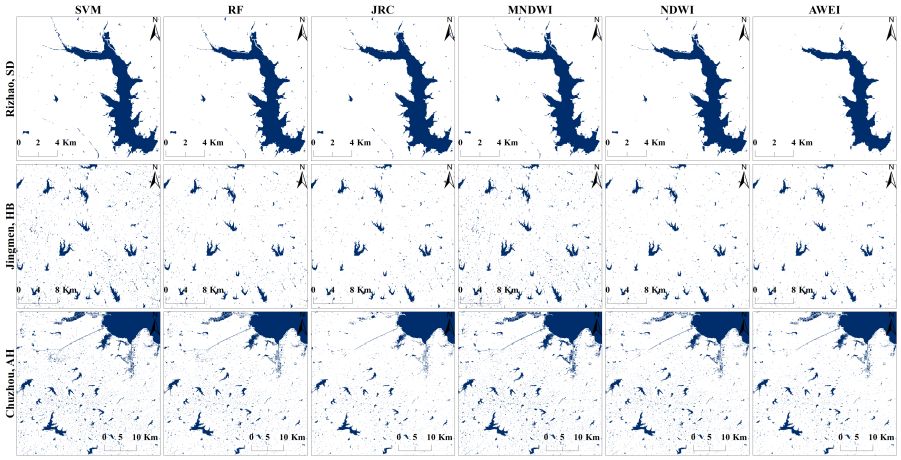

Fig. 6. Spatial distribution of water extraction results from multiple comparative methods.

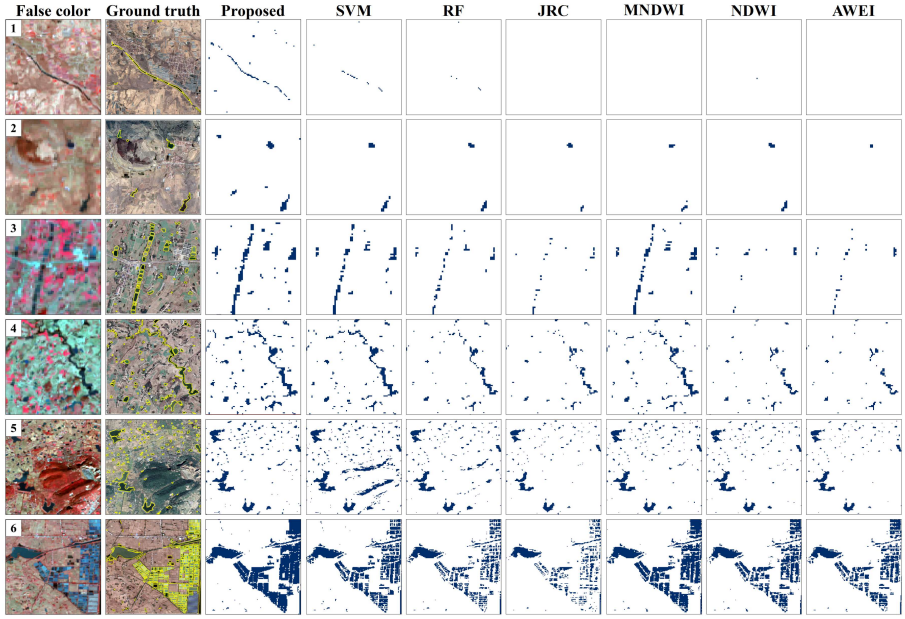

Fig. 7. Qualitative comparison of water extraction details in representative subregions.

contrast, the three WI methods and WF-MNDWI effectively suppressed shadow-induced misclassifications. In subregion 6, characterized by densely distributed aquaculture ponds, RF and JRC showed the largest discrepancies from ground truth, possibly due to mixed pixels caused by the narrow land strips between ponds; SVM, the three WI methods, and WF-MNDWI all extracted water bodies relatively completely, though differences in boundary sharpness were evident. It should be noted that the smoother boundaries in the WF-MNDWI results do not indicate methodological deficiencies; rather, they arise from its "water-containing pixel" extraction objective, which prioritizes the overall inclusion of water pixels in the binary outputs, reflecting a distinct treatment of mixed pixels compared with traditional methods.

Overall, the qualitative comparisons in Fig. 7 indicate that among the supervised classification methods, SVM outperforms RF, while JRC exhibits the poorest performance. Among the WI-based methods, MNDWI achieves the best overall extraction accuracy, with NDWI and AWEI showing comparable performance. Across all scenarios, WF-MNDWI consistently outperforms all comparative methods in both recovering the continuity of narrow river channels and detecting isolated SWBs, demonstrating superior comprehensive extraction capability.

C. Quantitative Accuracy Assessment

Based on the validation sample points (Fig. 3), four metrics—OA, UA, PA, and the $F1$ score—were employed to quantitatively evaluate the water extraction performance of WF-MNDWI and all comparative methods (Table IV). Overall, all methods demonstrated basic capabilities in water body detection, with OA values exceeding 80% across all study areas (Fig. 8). However, these methods exhibited notable

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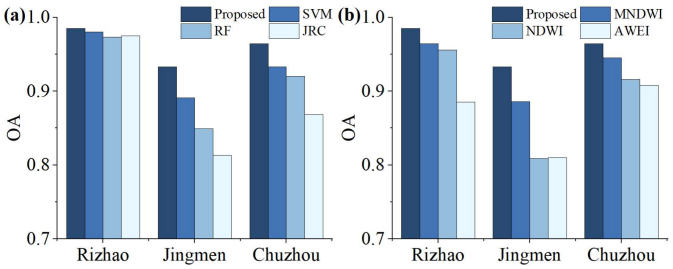

Fig. 8. Comparison of overall accuracy for water extraction by different methods. (a) Comparison between the proposed method and SVM, RF, and JRC; (b) comparison between the proposed method and MNDWI, NDWI, and AWEI.

TABLE IV

WATER EXTRACTION ACCURACIES FOR SELECTED STUDY AREAS IN COMPARISON TO DIFFERENT CLASSIFICATION METHODS

Study area	Method	OA(%)	UA(%)	PA(%)	F1-score
Rizhao, SD	Proposed	**98.20**	98.38	**97.98**	**0.9818**
	SVM	98.00	100.00	95.96	0.9794
	RF	97.30	100.00	94.55	0.9720
	JRC	97.50	99.79	95.15	0.9741
	MNDWI	96.40	100.00	92.73	0.9623
	NDWI	95.60	100.00	91.11	0.9535
	AWEI	88.50	100.00	76.77	0.8686
Jingmen, HB	Proposed	**93.30**	91.10	**91.34**	**0.9122**
	SVM	89.10	95.03	75.33	0.8404
	RF	84.90	97.92	61.68	0.7568
	JRC	81.30	98.50	51.71	0.6781
	MNDWI	88.60	95.25	73.75	0.8314
	NDWI	80.90	100.00	49.87	0.6655
	AWEI	81.00	98.97	50.66	0.6701
Chuzhou, AH	Proposed	**96.40**	98.70	**93.81**	**0.9619**
	SVM	93.30	99.29	86.80	0.9263
	RF	92.00	99.27	84.12	0.9107
	JRC	86.80	99.72	72.99	0.8429
	MNDWI	94.50	99.31	89.28	0.9403
	NDWI	91.60	99.02	83.51	0.9060
	AWEI	90.80	99.50	81.44	0.8957

differences in the trade-off between commission and omission errors.

Specifically, the supervised classification methods (SVM, RF) and the JRC dataset achieved very high UA values (>95%) but relatively low PA values. Similarly, the WI-based methods (MNDWI, NDWI, AWEI) also exhibited the common characteristic of high UA (>95%) but low PA (<93%). These results indicate that although the comparative methods effectively ensured the purity of the extracted water pixels (i.e., high precision), they suffered from insufficient completeness in detecting small and fragmented water bodies, primarily due to the spectral variability introduced by mixed pixels.

In sharp contrast, the proposed WF-MNDWI method achieved substantially higher PA values across all three study areas (Rizhao: 97.98%; Jingmen: 91.34%; Chuzhou: 93.81%), while maintaining excellent UA values (>91%). This result demonstrates the superior effectiveness of the WF-MNDWI framework in enhancing the recall of SWBs.

Ultimately, owing to its balanced performance in both PA and UA, WF-MNDWI achieved the highest OA and $F1$ scores across all study areas (Table IV, Fig. 8). Compared with the second-best method in each study area, WF-MNDWI further improved OA by 0.20%–4.20%. In summary, the quantitative evaluation results consistently confirm that the proposed WF-MNDWI method outperforms existing mainstream approaches in terms of water body extraction accuracy.

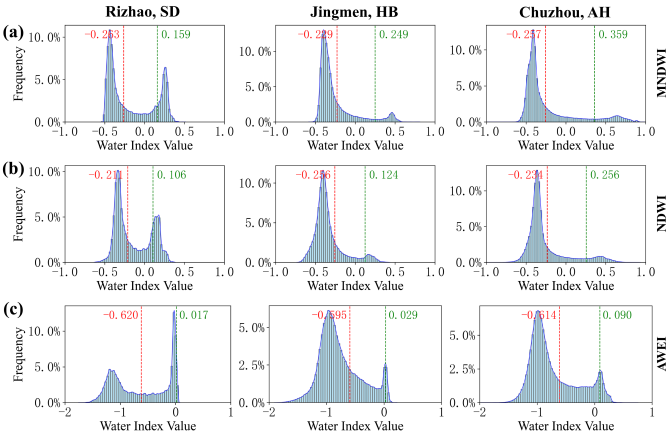

Fig. 9. Histogram distributions of different water indices and their automatically identified thresholds. (a)–(c) Correspond to MNDWI, NDWI, and AWEI, respectively.

V. DISCUSSIONS

A. Generality and Transferability of the WF-WI Framework

1) Performance Across Different Water Indices: The WF-WI framework exhibits a high degree of generalizability, allowing adaptation to various water indices for subpixel water extraction. To rigorously validate this property, three traditional water indices—MNDWI, NDWI, and AWEI—were selected to instantiate the WF-MNDWI, WF-NDWI, and WF-AWEI methods for comparative experiments. Fig. 9 presents the histogram distributions of the three water indices within each study area, along with the automatically derived T_{nonwater} and $T_{\text{pure-water}}$ thresholds, indicated by red and green dashed lines, respectively.

To ensure fairness in the comparison, all methods employed the identical set of land-cover endmember samples shown in Fig. 5. This design ensures that performance differences arise solely from the intrinsic characteristics of the water indices, rather than biases in endmember selection. For each method, water and land endmembers were automatically determined based on their WI values, strictly following the rules described in the methodology section.

Fig. 10(a) and (b) shows the water fraction maps and the corresponding binary water distribution results obtained with the recommended thresholds for each method. The quantitative accuracy evaluation results are summarized in Fig. 11. After applying the WF-WI framework, the OA accuracies achieved by the three implementations were similar, with only minor differences (WF-MNDWI > WF-AWEI > WF-NDWI). Notably, the framework substantially improved the OA accuracy of the baseline WIs, with the most significant gains observed for indices with relatively weak original performance (AWEI > NDWI > MNDWI). These findings demonstrate the strong capability of the WF-WI framework to compensate for the inherent limitations of weaker indices.

Although the WF-WI framework generally improves the performance of all baseline water indices, the inherent char-

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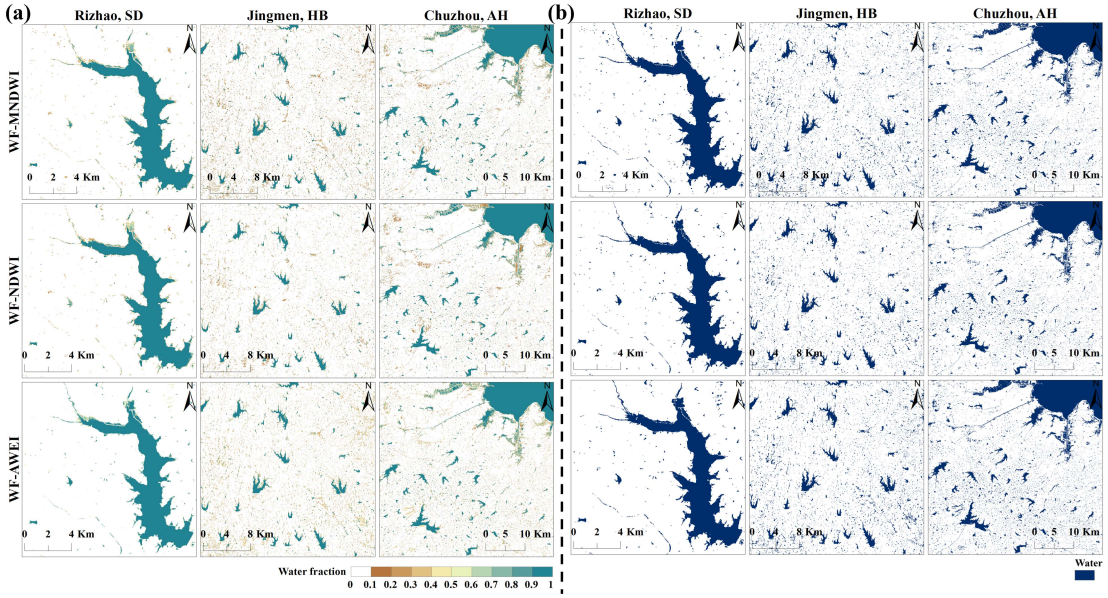

Fig. 10. Comparison of WF-WI framework outputs based on different water indices. (a) Water fraction. (b) Binary water.

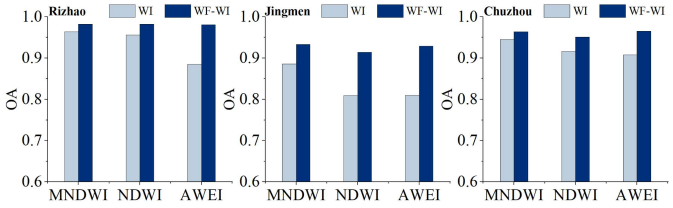

Fig. 11. Enhancement effect of the WF-WI framework on the extraction accuracy of different water indices.

acteristics of different indices in enhancing water signals and suppressing nonwater backgrounds lead to pronounced differences in fine-scale extraction performance. Fig. 12 provides a qualitative comparison for three representative subregions (top row: binary extraction results; bottom row: water fraction maps).

In the region with mountain shadows highlighted by yellow circles in Fig. 12(a), WF-MNDWI exhibits the strongest resistance to shadow-induced interference, with the fewest FPs. In contrast, WF-AWEI shows substantial false detections, while WF-NDWI partially mitigates shadow effects but misclassifies high-reflectance surfaces (red box). For narrow river extraction in Fig. 12(b) (yellow circle), WF-AWEI proves most effective. Regarding the high-reflectance surfaces (red box), WF-AWEI is least affected, followed by WF-MNDWI, while WF-NDWI is most susceptible to misclassification. The analysis of Fig. 12(c) further shows that WF-AWEI continues to perform best in extracting extremely narrow river networks (yellow circle). For aquaculture pond boundaries (red box), WF-NDWI yields the sharpest delineations, with WF-MNDWI following closely behind. It must be emphasized that the smoother boundaries observed in other results do not reflect methodological shortcomings; rather, they directly stem from the framework's design objective to extract subpixel "water-containing" pixels—narrow terrestrial levees may be too small

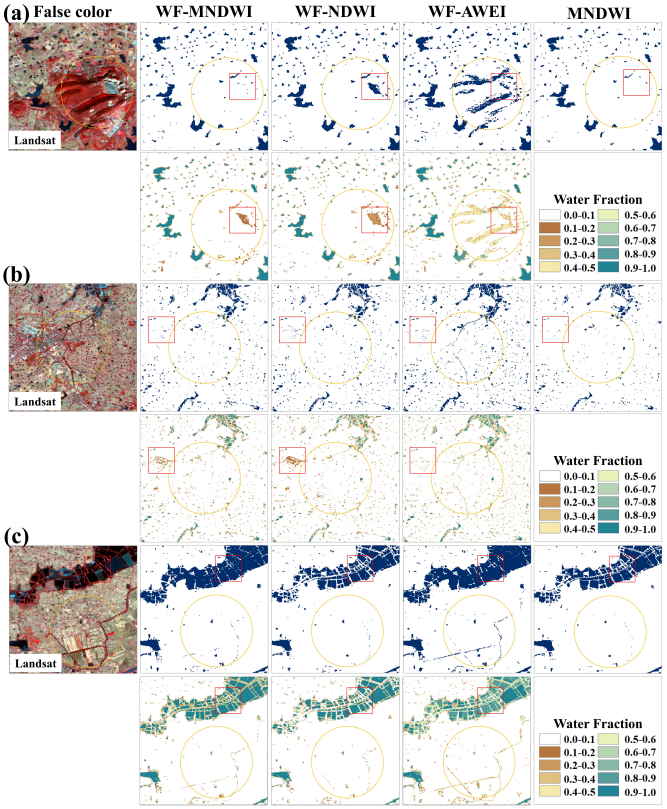

Fig. 12. Comparison of detailed extraction performance by WF-WI methods based on different water indices in representative scenarios. (a) Mountain shadow region, (b) narrow river region, and (c) aquaculture pond region.

to constitute a complete pixel and are thus reasonably classified as water.

In summary, the analysis reveals a performance trade-off among WF-WI implementations based on different indices, driven by the intrinsic strengths and weaknesses of the baseline WIs: WF-AWEI excels in detecting narrow rivers but is most

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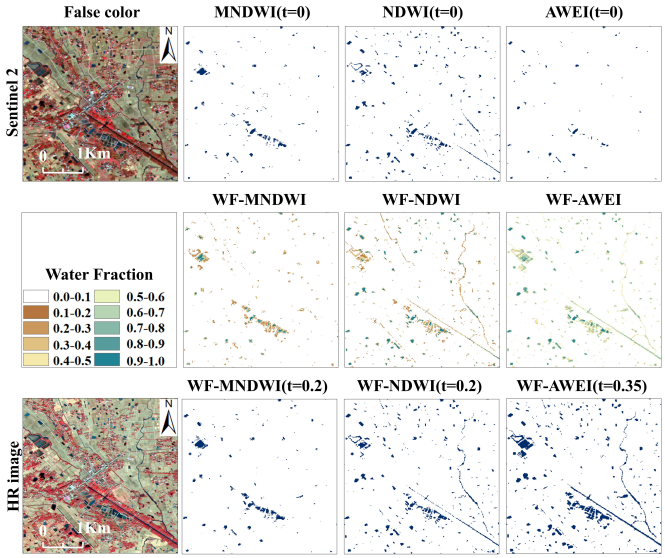

Fig. 13. Cross-sensor performance validation of the WF-WI framework on Sentinel-2 imagery.

vulnerable to mountain shadow interference; WF-MNDWI offers the most robust and reliable overall performance across diverse terrains with strong shadow-resistance; WF-NDWI, while slightly inferior in pure water detection and more prone to misclassifying bright surfaces, provides the sharpest water boundaries and, owing to its reliance solely on green and NIR bands, holds unique potential for extending the framework to other satellite sensors.

2) Transferability to Sentinel-2 Imagery: To evaluate the generalization capability and transferability of the WF-WI framework beyond the Landsat OLI sensor, we applied it to Sentinel-2 MSI imagery for validation. The visible-to-SWIR bands (B1, B2, B3, B4, B8, B11, B12) with spectral responses similar to those of Landsat were selected. For experimental consistency, the endmember sample set was kept identical to that used in the Landsat experiments, and the same recommended thresholds were adopted for binary water extraction.

Fig. 13 systematically compares the output results of the WF-WI framework applied to Sentinel-2 imagery using MNDWI, NDWI, and AWEI, respectively. The visual results indicate that the WF-WI framework can be successfully transferred to the Sentinel-2 sensor, consistently enhancing the extraction performance of all baseline water indices. Specifically, WF-AWEI demonstrates outstanding performance in maintaining the completeness of small water body extraction, a finding consistent with the Landsat experiments, further confirming the robustness of the WF-AWEI method for subpixel water extraction.

However, one noteworthy phenomenon is that WF-NDWI significantly outperforms WF-MNDWI on Sentinel-2 imagery—a result that contrasts with the findings based on Landsat data. Our preliminary analysis suggests that this discrepancy may be partly attributed to system-level differences between sensors, but more directly, it is likely related to the spatial resolution of the input bands. The MNDWI calculation incorporates the 20 m resolution B11 band (upsampled to 10 m), meaning its water fraction maps remain inherently constrained by the original resolution. In contrast, NDWI relies solely on the green and NIR bands, both natively at 10 m resolution, enabling WF-NDWI to produce cleaner and more detailed 10 m water fraction maps. This ultimately results in visually superior extraction performance compared with WF-MNDWI. The substantial quality differences among the baseline WI extraction results shown in the top row of Fig. 13 further support this interpretation.

In summary, this experiment demonstrates the strong cross-sensor transferability of the WF-WI framework. Although differences in spectral characteristics and spatial resolution among sensors may lead to variations in the relative performance rankings of different WIs, the core advantage of the framework—enhancing the water extraction capability of baseline WIs—has been consistently validated across multiple data sources.

B. Core Advantages and Mechanistic Analysis of the WF-WI Framework

Compared with traditional supervised classification methods and WI-based approaches, the proposed WF-WI framework demonstrates multiple technical advantages in extracting SWBs. Its superiority stems from its unique design philosophy and processing mechanisms, which can be summarized in three key aspects.

1) *Low Data Dependency and High Computational Efficiency:* Traditional supervised classification methods heavily rely on large volumes of high-quality labeled samples to train complex models, resulting in high computational costs and susceptibility to overfitting. In contrast, the WF-WI frame-work requires only a small set of representative land-cover spectral reflectances as input. Through a standardized work-flow of "initial classification → endmember adaptation → water fraction calculation," fully automated water extraction can be achieved. Compared with conventional linear spectral mixture analysis, which performs least-squares solutions across all pixels, the framework innovatively focuses on the subset of "potential water-containing mixed pixels" and employs water indices to construct a simplified decomposition model. This substantially reduces computational complexity and provides a practical solution for large-area, long-term dynamic monitoring of SWBs.
2) *Advanced Mixed-Pixel Processing Mechanism:* The core breakthrough of the framework lies in overcoming the inherent theoretical limitations of traditional water indices. Conventional indices, such as MNDWI, rely on a strong implicit assumption: the reflectance of water in the green band is always higher than that in the SWIR band. Fig. 14(a), based on a linear spectral mixture model simulation, demonstrates that when the proportion of high-reflectance land components in a mixed pixel exceeds a critical threshold, the SWIR reflectance may exceed the green reflectance, causing traditional water indices to fail (e.g., MNDWI <0), even if the actual water fraction is high (e.g., 0.8). Although

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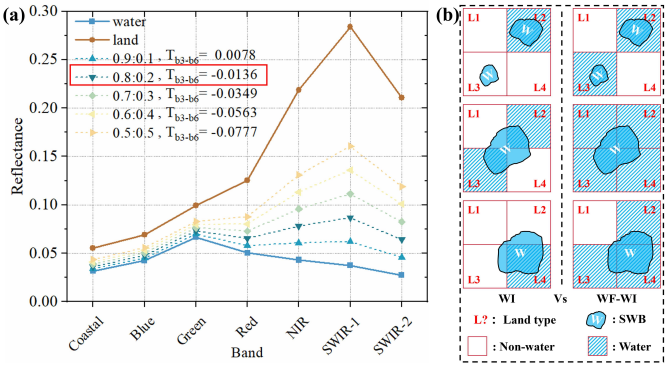

Fig. 14. Conceptual diagrams illustrating (a) mixed-pixel spectral simulation and (b) spatial partitioning of small water bodies.

lowering the threshold could theoretically mitigate this issue, the spectral complexity of land cover and water makes a universally optimal threshold infeasible, leaving traditional methods trapped in a trade-off between commission and omission errors. Fig. 14(b) conceptually illustrates the incompleteness and randomness of traditional threshold-based extraction results under various scenarios in which SWBs are split across pixel grids. In contrast, the WF-WI framework independently estimates water fraction for each pixel, fundamentally enhancing its interpretability of mixed pixels. Regardless of how water spans across pixels or mixes with land types, all water-containing pixels can theoretically be identified, ensuring much greater completeness in SWB extraction. Although absolute fraction accuracy may be affected by linear model assumptions and endmember variability, the robustness of binary water/nonwater decisions is far superior to conventional methods.

3) *Outstanding Framework Generalizability and Transferability:* This study has successfully validated the seamless transfer of the WF-WI framework between two mainstream medium-resolution sensors, Landsat OLI and Sentinel-2 MSI. Notably, because WF-NDWI relies solely on the green and NIR bands—two of the most universally available spectral bands—the framework exhibits significant application potential and can be readily extended to other multispectral satellite systems (e.g., China's GF series, Terra/Aqua MODIS). This establishes a solid foundation for developing operational water monitoring workflows at global scale and across multisource datasets.

C. Uncertainty Analysis

The proposed WF-WI framework is based on the linear spectral mixing model, whose core assumption is that the spectral signature of a sensor-observed pixel can be represented as a linear combination of the spectra of various endmembers. However, the complexity of real-world surface environments inevitably introduces multiple sources of uncertainty in practical applications, which are primarily reflected in the following aspects:

Uncertainty Due to Endmember Variability. The assumption of fixed endmembers is often difficult to satisfy in reality. The spectral signature of water endmembers dynamically varies under the influence of factors such as turbidity, water depth, and chlorophyll concentration; soil endmembers differ according to moisture, mineral composition, and surface roughness; vegetation endmembers are affected by phenological stage and health status. This spatiotemporal heterogeneity of endmembers leads to deviations between actual mixed pixels and the predefined "pure" endmembers, potentially resulting in systematic errors in water fraction estimation [53], [61].

Uncertainty Due to Illumination and Topographic Effects. Although normalized water indices possess some atmospheric correction capability, in regions with pronounced terrain relief, shadow effects can alter the apparent reflectance of surface features, causing spectral confusion between deep valleys, shaded areas, and regions with low water fractions, thus increasing the risk of FPs [34]. Additionally, the nonlinear optical properties of shallow water further violate the linear mixing assumption, exacerbating classification uncertainty [62].

Estimation instability under low signal-to-noise ratios. When the water fraction within a pixel is extremely low, the weak signal can easily be overwhelmed by background spectral variability and sensor noise, rendering fraction estimation highly sensitive to endmember values and potentially producing anomalous results below 0 or above 1 (constrained to 0–1 in this study). This instability is mainly manifested as: 1) omission, where SWBs are missed due to insufficient signal and 2) commission, where dark nonwater features are misclassified as water due to spectral similarity.

These uncertainties do not undermine the WF-WI framework but instead define its applicable scope and confidence intervals. Accordingly, this study adopts two mitigation strategies: 1) a dual-threshold preclassification to exclude clear nonwater areas and reduce FPs and 2) a spatially adaptive endmember selection to ease heterogeneity effects. For instability in low water fractions, users are advised to adjust thresholds based on precision needs to balance recall and accuracy in applications.

D. Limitations and Future Work

The WF-WI framework proposed in this study demonstrates excellent performance in the extraction of SWBs, yet two main limitations remain, providing directions for future improvement.

First, the framework exhibits insufficient automation and scalability. Currently, endmember extraction still relies on manual sample selection, which is difficult to meet the demands of large-scale or near-real-time monitoring. Recent studies have shown that automated training sample generation methods, combining multisensor optical imagery with machine learning, can significantly enhance the intelligence and efficiency of water and flood detection [63]. Therefore, in the future, the WF-WI framework could be deployed on cloud platforms such as GEE, and automated techniques of this kind could be adopted to construct a fully automated pipeline from endmember selection to water extraction, thereby improving the framework's scalability and operational potential.

Second, the framework is limited by data sources and applicable scenarios. The WF-WI framework primarily relies

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on optical imagery, whose availability and information quality can be constrained under cloudy, rainy, or complex terrain conditions. Recent studies have demonstrated that methods integrating SAR and optical data exhibit significant advantages in identifying SWBs in complex terrain [64]. Therefore, future work could explore the development of an SAR-optical collaborative extension of the WF-WI framework, leveraging the complementary information of optical water fractions and SAR backscatter characteristics to achieve higher accuracy and robustness in water mapping across diverse global environments.

VI. Conclusion

To address the issue of missed detection of SWBs caused by mixed pixels, this study proposes a subpixel water extraction framework, WF-WI, which integrates WI with the linear spectral mixing model to quantitatively estimate the water fraction within mixed pixels. Experiments based on Landsat 8/9 imagery in three study areas in China indicate that WF-MNDWI significantly outperforms SVM, RF, JRC, and traditional index-based methods in terms of $F1$ scores (0.9818, 0.9122, 0.9619) and OA, with an accuracy improvement of up to 4.20%. Cross-index and cross-sensor validation results demonstrate that WF-WI is compatible with MNDWI, NDWI, AWEI, and Sentinel-2 data, exhibiting good generality and transferability.

Different indices have specific advantages under particular scenarios: WF-MNDWI is robust in most cases and resistant to topographic shadow interference; WF-AWEI performs best for extracting narrow water bodies under shadow-free conditions; although WF-NDWI is affected by highly reflective surfaces, it still outperforms traditional indices. Methodologically, WF-WI enhances the detection of subpixel SWBs through a dual-threshold preclassification and spatially adaptive endmember selection strategy. Furthermore, by focusing on potential water-containing mixed pixels, the framework substantially reduces computational load, providing a novel technical pathway for rapid and automated large-scale monitoring of SWBs.

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